HOLOGRAPHIC TOPOLOGY ENGINE

System Testing Manual & Operator's Reference Guide — Software Release 2026.4

DOCUMENT ID	TOTAL TEST SUITES	SAMPLING RATE	COMPLIANCE
HTE-MAN-2026-V1	13 Selectable	5.0 - 10.0 Hz	APS / IEEE Standard

1. SYSTEM MANUAL & PHYSICS TEST SUITE SPECIFICATION

The **Scientific Testing Suite** provides standardized real-time telemetry verification, invariant sampling, and numeric convergence testing across 13 core and experimental physical system presets. Each benchmark evaluates spatial discretization, topological invariants, energy conservation, or entanglement dynamics under controlled thermal and non-Hermitian perturbations.

1. Base Topology Verification (*topologyLattice.html*)

CORE SUITE

Evaluates non-trivial Chern number invariants in discretized 2D/3D topological insulators under variable lattice deformations and gauge field perturbations.

Berry Curvature: $F_{xy}(k) = \partial_x A_y - \partial_y A_x$ | Chern Number: $C = (1 / 2\pi) \iint_{BZ} F_{xy}(k)\, dk_x dk_y$

Expected Results: Quantized integer Chern invariant $C \in \mathbb{Z}$ (target: $C = 1$) with spatial mesh resolution $N \geq 2500$ and numerical boundary drift $< 10^{-6}$.

2. Spacetime Curvature & Geodesics (*spacetimeMetric.html*)

CORE SUITE

Integrates ray-tracing geodesics through Schwarzschild and Kerr black hole metric tensors, measuring photon trajectory deflections and frame dragging.

Geodesic Equation: $d^2x^\mu/d\lambda^2 + \Gamma^\mu_{\alpha\beta}\,(dx^\alpha/d\lambda)(dx^\beta/d\lambda) = 0$

Expected Results: Stable photon sphere radius at $r = 3GM/c^2$; energy conservation parameter $E = -p_0$ invariant to within 10^{-5} over 100s execution.

3. Holographic Boundary Field (*boundaryField.html*)

CORE SUITE

Maps bulk Anti-de Sitter (AdS) scalar field fluctuations onto boundary Conformal Field Theory (CFT) energy-momentum correlation functions.

Bulk-Boundary Duality: $\langle O(x)\, O(y) \rangle = C_\Delta / |x - y|^{2\Delta}$ where $m^2L^2 = \Delta(\Delta - d)$

Expected Results: Power-law decay matching conformal dimension $\Delta = 3.0$ with radial cutoff boundary scaling deviation $< 0.02\%$.

1. SYSTEM MANUAL (CONTINUED — TESTS 4 TO 8)

4. Quantum Entanglement & Entropy (*entanglementEntropy.html*)

CORE SUITE

Computes Ryu-Takayanagi minimal surface areas in bulk spacetime to quantify boundary Von Neumann entanglement entropy.

Ryu-Takayanagi Formula: $S_A = Area(\gamma_A) / (4\ G_N^{(d+2)})$

Expected Results: Holographic area-law scaling $S_A \propto \partial A$ with logarithmic sub-leading corrections verified to SNR $> 42\ dB$.

5. Bulk Gravity Soliton Stability (*gravitySoliton.html*)

CORE SUITE

Evaluates non-singular Einstein-Yang-Mills soliton field configurations, testing non-linear stability against radial perturbations.

Soliton Mass Functional: $M = \int_0^\infty [\ (1/2)\ m'(r) + 4\pi\ r^2\ T_0^{\ 0}\ ]\ dr$

Expected Results: Asymptotically flat ADM mass profile M_{ADM} convergence with peak residual drift $< 1e\text{-}7$ over 120s runtime.

6. Non-Abelian Gauge Holonomy (*gaugeHolonomy.html*)

CORE SUITE

Tracks SU(2) / SU(3) gauge field parallel transport along closed lattice loops, extracting Wilson loop expectation values.

Wilson Loop: $W(C) = Tr\ [\ P\ exp(\ i\ \oint_C A_\mu\ dx^\mu\ )\ ]$

Expected Results: Quantized trace bounds $|W(C)| \leq N_c$ (for $SU(N_c)$) with unitarity conservation error $< 10^{-8}$.

7. Higher-Dimensional Bulk Projections (*bulkProjections.html*)

CORE SUITE

Projects 5D AdS hyper-surface geometries onto 3D visual slices using isometric embedding matrices.

Embedding Metric: $ds^2 = (L^2/z^2)\ [\ dz^2 + \eta_{ij}\ dx^i\ dx^j\ ]$

Expected Results: Smooth isometric projection with zero metric determinant singularities ($det(g) > 0$).

8. Anyonic Circuit Decoherence (*wishlist_anyonic_circuit*)

EXPERIMENTAL WISHLIST

Simulates Fibonacci non-Abelian anyon braiding under stochastic thermal noise ($15\ mK$) and flux jitter.

Braiding Phase: $R_{\tau\tau} = exp(i\ 3\pi\ /\ 5)$ | Fidelity: $F = \langle\psi_0|\ \rho_t\ |\psi_0\rangle$

Expected Results: Topological gate fidelity $F \geq 0.999$; leakage rate out of computational subspace < 0.001 .

1. SYSTEM MANUAL (CONTINUED — TESTS 9 TO 13)

9. Kibble-Zurek Criticality & Quench (*wishlist_kibble_zurek*)

EXPERIMENTAL WISHLIST

Executes continuous parameter quenches across 3D Ising topological critical points to evaluate defect density scaling.

Defect Density Scaling Law: $n_d \propto \tau_q^{-\nu d/(1+z\nu)}$

Expected Results: Universal power-law defect exponent $\nu d/(1+z\nu) \approx 0.75$ in 3D Ising class; non-negative defect count.

10. Non-Hermitian Skin Effect (*wishlist_non_hermitian*)

EXPERIMENTAL WISHLIST

Simulates dissipative open quantum systems, tracking bulk eigenmode collapse onto boundaries at non-Hermitian exceptional points.

Skin Localization Length: $\xi = 1/\ln|g/\gamma|$ | Point-Gap Winding: $W = (1/2\pi i) \oint dk\, \partial_k \ln \det[H(k) - E_0]$

Expected Results: Non-zero point-gap winding number $W = 1$ under open boundary conditions; finite complex energy spectrum.

11. AdS Black Hole Thermalization (*wishlist_ads_black_hole*)

EXPERIMENTAL WISHLIST

Models scalar field infall and horizon formation in AdS5 spacetime, tracking dual CFT Page curve entropy evolution.

Scrambling Lyapunov Bound: $\lambda_L \leq 2\pi k_B T/\hbar$

Expected Results: Lyapunov exponent saturating chaos bound $\lambda_L \to 2\pi T$; Page curve turnover at horizon formation time t_p .

12. PEPS Tensor Network Ingestion (*wishlist_peps_tensor*)

EXPERIMENTAL WISHLIST

Ingests Projected Entangled Pair States (PEPS) to extract topological entanglement entropy γ in Fractional Quantum Hall states.

Topological Entanglement Entropy: $S(L) = \alpha L - \gamma$ where $\gamma = \ln D$

Expected Results: Quantized universal constant $\gamma = \ln(\sqrt{3}) \approx 0.549$ for $\nu = 1/3$ Laughlin state with bond dimension $D = 8$.

13. Non-Selectable Config System (*test_suite_configs/*)

SYSTEM CORE

Underlying schema definitions (`test_config.schema.json`) and configuration loader that enforce runtime parameters, sampling boundaries, and automated pass/fail validation logic across all tests.

2. SCIENTIFIC TESTING SUITE OPERATOR'S GUIDE

To ensure clean, reproducible telemetry ingestion and prevent memory leaks or temporal jitter during high-frequency execution ($5 - 10\ Hz$), operators must follow standard execution protocols.

1 Step-by-Step Clean Operation Protocol

1. Pre-Flight Inspection: Verify GPU hardware acceleration is active (`webgl2` or `webgpu`). Ensure no secondary browser tabs consume engine worker threads.
2. Preset Selection: Launch the suite drawer, select the desired scenario (1–13), and click *"Load Configuration"*. Confirm parameters match target physics.
3. Capture Execution: Initiate automated capture. Do not resize or minimize the window during the sampling interval to maintain strict 200ms frame spacing.
4. Validation Verification: Ensure all telemetry metrics remain within green validation bounds ($SNR > 35\ dB$, $singularity < 10^{-6}$).

3. PUBLISHING ASSISTANT & `ANALYZE_TELEMETRY.IPYNB`

The **Publishing Assistant** converts raw telemetry JSON logs into publication-ready scientific manuscripts and APS-compliant figures without manual post-processing.

JAX Ingestion & Estimators

The `analyze_telemetry.ipynb` notebook utilizes `jax.numpy` for zero-copy memory mapping. It applies `jax.jit` compiled functions to evaluate finite-difference derivatives dN/dt , temporal jitter spectra, and asymptotic scaling exponents in seconds.

APS Journal Formatting

Generates vector figures (`.pdf`/`.png`) formatted for *Physical Review Letters / B* (3.375-inch single column width, 300 DPI, Computer Modern serif typography, and explicit panel designations **A**, **B**).

Post-Capture Integration: After completing a capture run in the Scientific Testing Suite, operators can click **"Export Telemetry Audit (JSON)"** and immediately execute `analyze_telemetry.ipynb` or trigger the LaTeX paper builder to obtain submission-ready deliverables.